Nivel 1

Descarga los archivos CSV, estudiales y diseña una base de datos con un esquema de estrella que contenga al menos 4 tablas de las que puedas realizar las siguientes consultas:

```sql
CREATE DATABASE IF NOT EXISTS world_transactions;
```

Output

Action Output

#	Time	Action	Message
1	22:05:38	CREATE DATABASE IF NOT EXISTS world_transactions	1 row(s) affected

```sql
CREATE TABLE transactions (
id VARCHAR(150) PRIMARY KEY,
card_id VARCHAR(100),
business_id VARCHAR(100),
timestamp DATETIME,
amount DECIMAL(10,2),
declined TINYINT,
product_ids VARCHAR(100),
user_id INT,
lat DECIMAL(30,15),
longitude DECIMAL(30,15)
);
```

Output

Action Output

#	Time	Action	Message
1	10:26:54	CREATE TABLE transactions (id VARCHAR(150) PRIMARY KEY, card_id VARCHAR(100), bu...	0 row(s) affected

```sql
LOAD DATA INFILE 'C:\\mysql-uploads\\transactions.csv'
INTO TABLE transactions
FIELDS TERMINATED BY ';'
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 LINES;
```

Output

Action Output

#	Time	Action
1	11:46:17	LOAD DATA INFILE 'C:\\mysql-uploads\\transactions.csv' INTO TABLE transactions FIELDS ...

```sql
CREATE TABLE users (
id INT PRIMARY KEY,
name VARCHAR(100),
surname VARCHAR(100),
phone VARCHAR(100),
email VARCHAR(150),
birth_date VARCHAR(50),
country VARCHAR(150),
city VARCHAR(150),
postal_code VARCHAR(50),
address VARCHAR(150));
```

Output

Action Output

#	Time	Action	Message
1	18:57:06	CREATE TABLE users (id INT PRIMARY KEY, name VARCHAR(100), surname VARCHAR(10...	0 row(s) affecte

```sql
LOAD DATA INFILE 'C:\\mysql-uploads\\american_users.csv'
INTO TABLE users
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES
(id, name, surname, phone, email, birth_date, country, city, postal_code, address);
```

Output

Action Output

#	Time	Action	Message
1	18:57:43	LOAD DATA INFILE 'C:\\mysql-uploads\\american_users.csv' INTO TABLE users FIELDS TE...	1010 row(s) affected Reco

```sql
LOAD DATA INFILE 'C:\\mysql-uploads\\european_users.csv'
INTO TABLE users
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES
(id, name, surname, phone, email, birth_date, country, city, postal_code, address);
```

Action Output

#	Time	Action	Message
1	18:58:20	LOAD DATA INFILE 'C:\\mysql-uploads\\european_users.csv' INTO TABLE users FIELDS TE...	3990 row(s) affect

```sql
CREATE TABLE credit_cards (
id VARCHAR(100) PRIMARY KEY,
user_id INT,
iban VARCHAR(100),
pan VARCHAR(100),
pin CHAR(4),
cvv CHAR(4),
track1 VARCHAR(255),
track2 VARCHAR(255),
expiring_date VARCHAR(50)
);
```

Output

Action Output

#	Time	Action
1	18:59:12	CREATE TABLE credit_cards (id VARCHAR(100) PRIMARY KEY, user_id INT, iban VARCHA...

```sql
LOAD DATA INFILE 'C:\\mysql-uploads\\credit_cards.csv'
INTO TABLE credit_cards
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES;
```

Output

Action Output

#	Time	Action	Message
1	18:59:48	LOAD DATA INFILE 'C:\\mysql-uploads\\credit_cards.csv' INTO TABLE credit_cards FIELDS ...	5000 row(s) affected

```sql
CREATE TABLE companies (
company_id VARCHAR(100) PRIMARY KEY,
company_name VARCHAR(255),
phone VARCHAR(100),
email VARCHAR(150),
country VARCHAR(150),
website VARCHAR(250));
```

Output

Action Output

#	Time	Action	Message
1	19:00:32	CREATE TABLE companies (company_id VARCHAR(100) PRIMARY KEY, company_name V...	0 row(s) aff

```sql
LOAD DATA INFILE 'C:\\mysql-uploads\\companies.csv'
INTO TABLE companies
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 LINES;
```

Output

Action Output

#	Time	Action	Message
1	19:02:01	LOAD DATA INFILE 'C:\\mysql-uploads\\companies.csv' INTO TABLE companies FIELDS TE...	100 row(s) affectec

```sql
ALTER TABLE transactions
ADD CONSTRAINT FK_cards_transactions
FOREIGN KEY (card_id) REFERENCES credit_cards(id);
```

Output

Action Output

#	Time	Action	Message
1	19:14:30	ALTER TABLE transactions ADD CONSTRAINT FK_cards_transactions FOREIGN KEY (card_i...	100000 row

```sql
ALTER TABLE transactions
ADD CONSTRAINT FK_users_transactions
FOREIGN KEY (user_id) REFERENCES users(id);
```

Output

Action Output

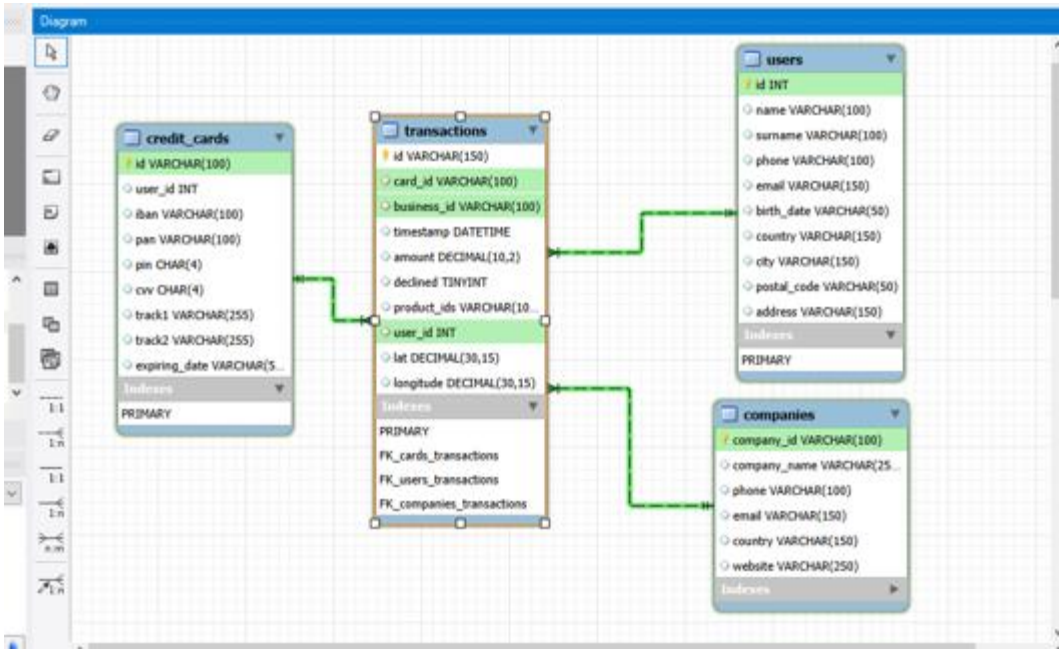
#	Time	Action	Message
1	19:18:17	ALTER TABLE transactions ADD CONSTRAINT FK_users_transactions FOREIGN KEY (user_i...	100000 row

```sql
ALTER TABLE transactions
ADD CONSTRAINT FK_companies_transactions
FOREIGN KEY (business_id) REFERENCES companies(company_id);
```

Output

Action Output

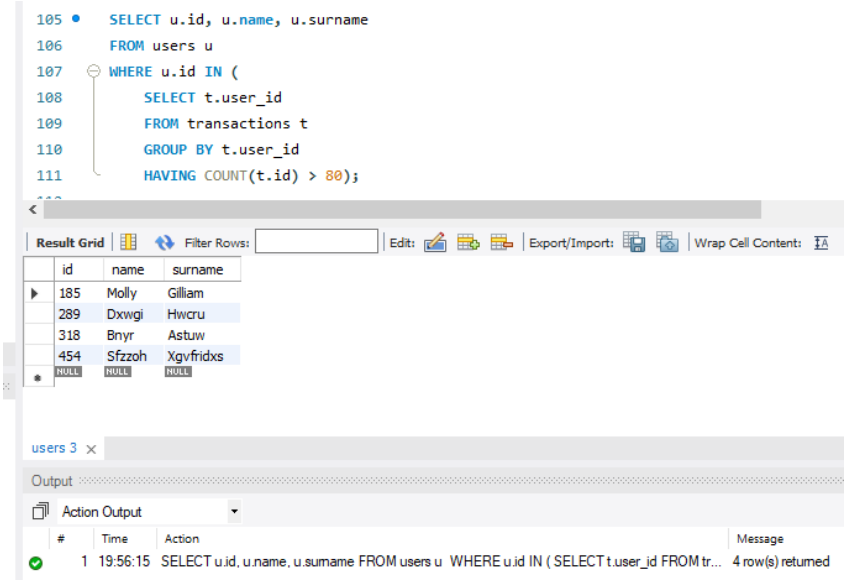
#	Time	Action	Message
1	19:20:36	ALTER TABLE transactions ADD CONSTRAINT FK_companies_transactions FOREIGN KEY (...	100000 row(s) affected



He creado la base de datos y cargado los datos, después de haber solucionado lo de LOAD DATA INFILE me he dado cuenta que no podía cargar algunos archivos y allí al final vi la diferencia de salto de página entre Windows y Linux. Despuès que relacionado las dimensiones con la tabla de hechos a través de fk constraints.

Ejercicio 1

Realiza una subconsulta que muestre a todos los usuarios con más de 80 transacciones utilizando al menos 2 tablas.



```
SELECT u.id, u.name, u.surname
FROM users u
WHERE u.id IN (
    SELECT t.user_id
    FROM transactions t
    GROUP BY t.user_id
    HAVING COUNT(t.id) > 80);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	name	surname
185	Molly	Gilliam
289	Dxwgi	Hwcru
318	Bnyr	Astuw
454	Sfzzoh	Xgvfridxs
NULL	NULL	NULL

users 3

Output

Action Output

#	Time	Action	Message
1	19:56:15	SELECT u.id, u.name, u.surname FROM users u WHERE u.id IN (SELECT t.user_id FROM tr...	4 row(s) returned

Primero he escrito una consulta que devolviera user id de transactions con màs de 80 transacciones, consulta que he utilizado para encontrar nombre y apellido de la tabla users.

Ejercicio 2

Muestra la media de amount por IBAN de las tarjetas de crédito en la compañía Donec Ltd., utiliza por lo menos 2 tablas.

En un primer momento he realizado esta consulta utilizando una subconsulta, después de haberla realizada solo con join me he dado cuenta que la segunda era màs veloz. En los outputs de abajo se puede notar la diferencia entre las dos.

```
SELECT cc.iban, ROUND(AVG(t.amount), 2) AS average_amount
FROM transactions t
JOIN credit_cards cc
ON t.card_id = cc.id
JOIN companies c
ON t.business_id = c.company_id
WHERE c.company_name = 'Donec Ltd'
GROUP BY cc.iban
ORDER BY cc.iban;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

iban	average_amount
AD2777204763277722050982	214.48
AE491827142302887266369	262.12
AL65255766650838829611407660	169.96
AT461138859877579187	303.39
AT658218806585843788	57.25
AZ0324899373398740738648927 1	382.86

Result 2

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	16:26:35	SELECT cc.iban, ROUND(AVG(t.amount), 2) AS average_amount FROM transactions t JOIN ...	371 row(s) returned	0.344 sec / 0.000 se
2	16:27:13	SELECT cc.iban, ROUND(AVG(t.amount), 2) AS average_amount FROM transactions t JOIN ...	371 row(s) returned	0.031 sec / 0.000 se

Aquì había hecho una subconsulta pero me di cuenta que tardaba bastante. Entonces cambiè utilizando joins y vi que resultaba bastante màs ágil.

Nivel 2

Crea una nueva tabla que refleje el estado de las tarjetas de crédito basado en si las tres últimas transacciones han sido declinadas entonces es inactivo, si al menos una no es rechazada entonces es activo

Con lo que tenía que modificar datos sin WHERE y sin LIMIT el programa no me dejaba, tuve que desactivar sql safe updates y reactivarlo después de haber modificado los datos que necesitaba cambiar.

```
# Nivel 2
# Crea una nueva tabla que refleje el estado de las tarjetas de crédito bas
# han sido declinadas entonces es inactivo, si al menos una no es rechazada

SET SQL_SAFE_UPDATES = 0;
```

Output

Action Output

#	Time	Action	M
1	12:10:21	SET SQL_SAFE_UPDATES = 0	0 r

```
SET SQL_SAFE_UPDATES = 0;

UPDATE credit_cards
SET expiring_date = STR_TO_DATE(expiring_date, '%m/%d/%y');
```

Output

Action Output

#	Time	Action	Message
1	12:11:09	UPDATE credit_cards SET expiring_date = STR_TO_DATE(expiring_date, '%m/%d/%y')	5000 row(s) affec

```
ALTER TABLE credit_cards
MODIFY expiring_date DATETIME;
```

Output

Action Output

#	Time	Action	Message
1	13:21:12	ALTER TABLE credit_cards MODIFY expiring_date DATETIME	5000 row(s) a

```
UPDATE users
SET birth_date = STR_TO_DATE(birth_date, '%b %d, %Y');
```

Output

Action Output

#	Time	Action	Message
1	12:30:32	UPDATE users SET birth_date = STR_TO_DATE(birth_date, '%b %d, %Y')	5000 row(s) a

```
UPDATE users
SET birth_date = STR_TO_DATE(birth_date, '%b %d, %Y');

ALTER TABLE users
MODIFY birth_date DATETIME;
```

Output

Action Output

#	Time	Action	Message
1	12:34:36	ALTER TABLE users MODIFY birth_date DATETIME	5000 row(s) aff

```sql
ALTER TABLE users
MODIFY birth_date DATETIME;

SET SQL_SAFE_UPDATES = 1;
```

Output

Action Output

#	Time	Action
1	12:35:29	SET SQL_SAFE_UPDATES = 1

```sql
ALTER TABLE transactions
RENAME COLUMN timestamp to date_time;
```

Output

Action Output

#	Time	Action
1	17:40:22	ALTER TABLE transactions RENAME COLUMN timestamp to date_time

Aquì cambiè el nombre, no se si hacìa falta pero no me gustaba tener una columna con el nombre de una función.

```sql
CREATE TABLE IF NOT EXISTS credit_card_status (
id INT AUTO_INCREMENT PRIMARY KEY,
credit_card_id VARCHAR(100),
card_status ENUM('active', 'inactive'));
```

Output

Action Output

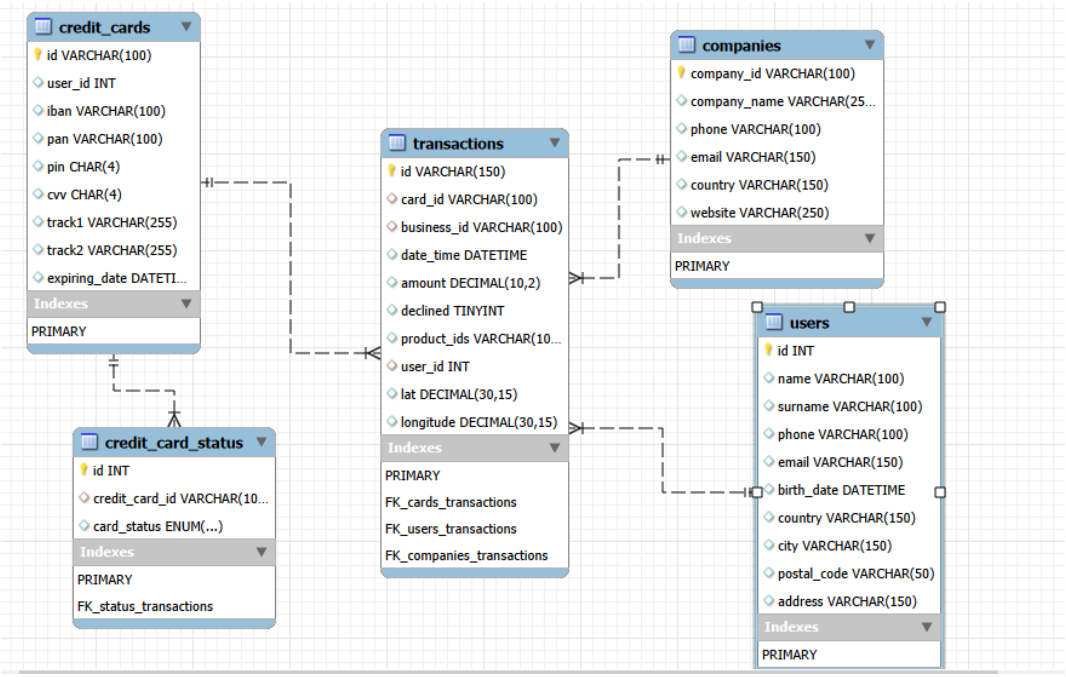
#	Time	Action	M
1	17:42:05	CREATE TABLE IF NOT EXISTS credit_card_status (id INT AUTO_INCREMENT PRIMARY K...	0 r

```sql
ALTER TABLE credit_card_status
ADD CONSTRAINT fk_status_card
FOREIGN KEY (credit_card_id) REFERENCES credit_cards(id);
```

Output

Action Output

#	Time	Action
1	17:44:25	ALTER TABLE credit_card_status ADD CONSTRAINT fk_status_card FOREIGN KEY (credit_...




```sql
SELECT card_id,
        CASE
          WHEN SUM(declined) = 3 THEN 'inactive'
          ELSE 'active'
        END AS card_status
FROM (
      SELECT card_id,
         date_time,
         declined,
         ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY date_time DESC ) AS rn
      FROM transactions) AS ranking
WHERE rn <= 3
GROUP BY card_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

card_id	card_status
CcS-4857	active
CcS-4858	active
CcS-4859	active
CcS-4860	active
CcS-4861	active

Result 7

Output

Action Output

#	Time	Action	Message
1	15:49:33	SELECT card_id, CASE WHEN SUM(declined) = 3 THEN 'inactive' ELSE 'activ...	5000 row(s) return

Aquì primero hice una consulta que devolvía id, fecha, si declined o no y una columna nueva que numerara las filas según orden cronològico descendente. La ultilizè como subconsulta para una consulta que me devolviera id y, según número de declined (3 ùltimos declined = inactiva) status de activa o inactiva en la nueva columna card status, limitando a las primeras tres de l'orden cronològico descendente, agrupando por id.

```
172 •  INSERT INTO credit_card_status(credit_card_id, card_status)
173    SELECT card_id,
174          CASE
175            WHEN SUM(declined) = 3 THEN 'inactive'
176            ELSE 'active'
177          END AS card_status
178    FROM (
179        SELECT card_id,
180           date_time,
181           declined,
182           ROW_NUMBER() OVER (PARTITION BY card_id ORDER BY date_time DESC ) AS rn
183        FROM transactions) AS ranking
184    WHERE rn <= 3
185    GROUP BY card_id;
186
187
188
189
190
191
192
```

Output

Action Output

#	Time	Action	Message
1	15:52:53	INSERT INTO credit_card_status(credit_card_id, card_status) SELECT card_id, CASE ...	5000 row(s) affected Records: 5000

Puse la consulta en un insert into, especificando las columnas porqè, siendo la tabla de 3 columnas totales (también hay id) sin esto me fallaba.

Partiendo de esta tabla responde: # Ejercicio 1 ¿Cuántas tarjetas están activas?

```
188 •  SELECT COUNT(*) AS active_cards
189    FROM credit_card_status
190    WHERE card_status = 'active';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

active_cards
4995

Result 9

Output

Action Output

#	Time	Action	Messa
1	15:54:43	SELECT COUNT(*) AS active_cards FROM credit_card_status WHERE card_status = 'active'	1 row(s

Una vez que tenía lista la tabla, solo hico falta hacer un count con condición de status activo.

Nivel 3

Crea una tabla con la que podamos unir los datos del nuevo archivo products.csv con la base de datos creada, teniendo en cuenta que desde transaction tienes product_ids.

```
CREATE TABLE IF NOT EXISTS products (
id INT PRIMARY KEY AUTO_INCREMENT,
product_name VARCHAR(250),
price VARCHAR(50),
colour VARCHAR(100),
weight DECIMAL(10,2),
warehouse_id VARCHAR(100)
);
```

Output

Action Output

#	Time	Action	M
1	16:28:32	CREATE TABLE IF NOT EXISTS products (id INT PRIMARY KEY AUTO_INCREMENT, prod...	0

```
LOAD DATA INFILE 'C:\\mysql-uploads\\products.csv'
INTO TABLE products
FIELDS TERMINATED BY ','
OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 LINES;
```

Output

Action Output

#	Time	Action	Message
1	16:31:24	LOAD DATA INFILE 'C:\\mysql-uploads\\products.csv' INTO TABLE products FIELDS TERMI...	100 row(s) affected

Aquì no intendìa bien como relacionar la tabla a las demás pero me di cuenta que el ejercicio no pedìa esto, entonces utilizè joins.

Ejercicio 1 Necesitamos conocer el número de veces que se ha vendido cada producto.

```sql
SELECT
    p.id,
    p.product_name,
    COUNT(*) AS times_sold
FROM products p
JOIN transactions t
ON FIND_IN_SET(
        p.id,
        REPLACE(t.product_ids, ' ', '')
    ) > 0
GROUP BY p.id, p.product_name
ORDER BY times_sold DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	product_name	times_sold
52	riverlands the duel	2654
29	Tully maester Tarly	2635
21	duel Direwolf	2609
16	the duel warden	2608
66	mustafar jinn	2601
87	sith Jade	2598

Result 9

Output

Action Output

#	Time	Action	Message
1	16:12:48	SELECT p.id, p.product_name, COUNT(*) AS times_sold FROM products p JOIN trans...	100 row(s) returned

Fin in set para encontrar el id en la lista de ids, >0 porquè los registros devuelven 0 si no hay coincidencias.

Solucionado con replace ya que el código no tenía cuenta de los espacios entre ids del csv, lo que supuso un problema.